

Content Hub Migration Plan

CHT Platform + Content Hub Producer (formerly MediaHub)

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Canonical migration runbook: target state, phased rollout, data port, serverless sync, and unified admin on contenthub.communityhealth.media/admin. Markdown source: [docs/contenthub-migration-plan.md](#)

1. Executive Summary

Migrate from the EC2 monolith MediaHub to: (1) Content Hub consumer + admin SPA on CloudFront/S3, (2) Content Hub producer API (contenthub-api) with serverless sync jobs, (3) two separate databases - CHT Aurora Global (platform) and producer Aurora Global in prod / RDS in dev.

Round 1 dev goal

- Producer API on ECS dev + RDS data port
- CHT dev wired via X-API-Key to `/api/public/*`
- First serverless sync Lambdas (tagging + cache clear)
- Admin shell at `/admin` on Content Hub (group-gated)

End state

- No EC2 monolith, no ECS scheduler worker, no MediaHub-hosted admin UI
- No GoTrue end-user auth on producer hostnames
- All sync/cron on EventBridge Scheduler + Lambda (+ SQS for long jobs)

2. Target State Architecture

```

CloudFront -> S3 contenthub.communityhealth.media (/ consumer, /admin CHM staff)
    |
    v
ALB -> cht-platform-backend (NestJS ECS) -> CHT Aurora Global (writer ue1, reader ue2)
    |           ElastiCache cht:catalog:*  Cognito  SQS
    |
    | X-API-Key (server only)
    v
ALB -> contenthub-api (FastAPI ECS) -> Producer Aurora Global (prod) / RDS (dev)
    ^
    |
EventBridge Scheduler -> Lambda / SQS -> sync, tagging, HCP intel
    |
    +-> POST /internal/cache/catalog/clear -> CHT backend

mediahub.{dev.,}communityhealth.media = APIs only (/api/public, /api/admin/studio, /webhook)
Post-MVP: mediahub-render (GPU + S3), mediahub-reports (SQS)

```

2.1 Services at target

Service	Dev	Prod	Notes
cht-platform-backend	ECS	ECS scale	Consumer + platform admin
contenthub-api	ECS	ECS scale	Public + studio APIs
Sync Lambdas	Yes	Yes	Replaces ECS worker
SQS + DLQ	Yes	Yes	Long platform sync
mediahub-render	-	Post-MVP	GPU FFmpeg
MediaHub Next.js admin	Bridge	Retired	Use /admin on Content Hub

2.2 Databases

Database	Dev	Prod	Owns
CHT platform	RDS/small Aurora	Aurora Global	Users, CME, client_ids
Content Hub producer	RDS Single-AZ	Aurora Global	Clips, tags, KOLs, HCP intel

Rule: cht-platform-backend NEVER connects to producer DB. HTTP only.

3. Locked Decisions

Area	Decision
Admin UI	contenthub.communityhealth.media/admin - same SPA, same session
Admin routes	/admin/platform/* (CHT) and /admin/studio/* (producer)
Cognito groups	chm-admin, chm-editor, chm-viewer - no mediahub-admin
Client scope	client_ids on CHT Aurora user; studio API enforces
HCP Intel admin	/admin/studio/hcp-intel/* on producer until future migration
CHT catalog	X-API-Key on /api/public/* (server-side only)
Admin studio	Cognito session JWT on /api/admin/studio/*
Producer prod DB	Aurora Global (ue1 writer, ue2 reader)
Producer dev DB	RDS Postgres instance
Sync jobs	Serverless: EventBridge + Lambda + SQS
shoot_tag_pipeline	Dormant - do not migrate until product decision

4. Auth Model

Actor	Target	Auth
End users	Consumer app	Cognito session cookie
CHM staff	/admin platform + studio	Same session + chm-* groups
CHT backend	/api/public/*	X-API-Key only
Admin SPA	/api/admin/studio/*	Session JWT + groups + client_ids
Sync Lambda	Producer DB	IAM VPC role
Sync Lambda	CHT cache clear	INTERNAL_CACHE_SECRET
ops-console	/webhook/sync	WEBHOOK_API_KEY

Cognito group mapping (legacy MediaHub roles)

Legacy role	Target group	Access
superadmin	chm-admin	Full platform + studio
admin	chm-admin	Full platform + studio
editor	chm-editor	Studio, scoped by client_ids
viewer	chm-viewer	Read-only studio

5. Admin UI (Content Hub)

CHM admins enter via Admin link in consumer app (visible only for chm-admin/editor/viewer). Same origin, same httpOnly session cookie. Never bookmark mediahub.* for UI.

Platform routes (/admin/platform/*) -> cht-platform-backend

- /admin/platform/users - Cognito-linked users, groups
- /admin/platform/access - until groups replace
- Future: CME, honorarium, surveys

Studio routes (/admin/studio/*) -> contenthub-api

- /admin/studio/content - tag editor (Domain 5)
- /admin/studio/analytics, /clients, /campaigns
- /admin/studio/clipper, /conversations, /render (Domain 6)
- /admin/studio/hcp-intel/* (Domain 9, producer DB today)
- /admin/studio/reports, /knowledge-base, /integrations

Studio API prefix (target)

All producer admin HTTP routes: /api/admin/studio/{domain}/...

- conversations, render, tags, analytics, clients, campaigns
- templates, hcp-intel, reports, knowledge-base

6. Domain Disposition (Producer Codebase)

Domain 5 - Content tagging

- post_tagger: shared lib / Lambda layer (foundation)
- playlist_doctor_tagger + post_tagging: Lambda cron
- tag_editor -> /api/admin/studio/tags

- shoot_tag_* : dormant, do not schedule

Domain 6 - Video / clip pipeline

- transcription, video_processor: SQS + Lambda/Fargate (post-R1)
- render_engine: mediahub-render ECS GPU + S3 (post-MVP)
- conversations, render routers -> /api/admin/studio/*
- auto_segmenter: retire (zero callers)

Domain 7 - Cross-cutting

- public_api -> /api/public/* (shrinks over time)
- scheduler -> EventBridge + Lambda (retire ECS worker)
- redis_store -> retired (CHT ElastiCache only)
- users, access_requests on producer -> retire (Cognito + CHT platform)

7. Serverless Sync Catalog

Replace APScheduler in ECS worker with EventBridge Scheduler.

Job	Schedule	Executor	Cache clear
Platform sync YT/LI/X/FB/IG	12h	SQS+Lambda	Yes
Platform stats	12h	SQS+Lambda	Yes
post_tagging	12h	Lambda	Yes
playlist_doctor_tagger	04:30 UTC	Lambda	Yes
ai_summaries	12h	SQS+Lambda	Yes
metric_snapshots	12h	Lambda	No
linkedin_* jobs	various	Lambda	No
hcp_intel_poll	30m	SQS+Lambda	No
hcp_intel backfill/resolve	cron	Lambda	No
kol_cache_warm	-	DROP	-
shoot_tag_pipeline	-	DO NOT MIGRATE	-

- Idempotent handlers; reserved_concurrency=1 on tagging jobs
- DLQ per queue; VPC access to producer DB; Secrets Manager for credentials
- cache_clear Lambda invoked after successful writes

8. Data Migration

8.1 Round 1 tables (producer dev RDS)

clips, posts, shoots, kols, kol_groups, kol_group_members, playlist_tags, hcps, hcp_signals, tag_audit, tag_proposal, tag vocabulary tables. Exclude/sanitize: users, client_users, auth PII.

8.2 Procedure

- Terraform producer dev RDS; alembic upgrade head

- pg_dump catalog subset from EC2 Postgres
- pg_restore to dev RDS; validate row counts (tags, doctor:*, KOLs, transcripts)
- Run tagging Lambdas once; CHT dev smoke via API key

8.3 Prod cutover

- pg_dump EC2 -> producer Aurora Global primary
- Parallel EC2 + new stack 48h; CHT MEDIAHUB_BASE_URL cutover
- Retire EC2; see mediahub-platform-cutover.md

9. Migration Phases

Phase 0: Prerequisites (parallel CHT Cognito Phase 2)

- 0.1 CHT: Cognito pools stable; end users off GoTrue
- 0.2 CHT: Groups chm-admin, chm-editor, chm-viewer in Terraform
- 0.3 Both: PUBLIC_API_KEY / MEDIAHUB_API_KEY in Secrets Manager
- 0.4 Producer: Block end-user login on legacy MediaHub UI
- 0.5 CHT: /internal/cache/catalog/clear + Redis cht:catalog:*

Phase 1: Dev infrastructure - ROUND 1 DEV

- 1.1 Terraform: producer dev RDS + contenthub-api ECS + ALB
- 1.2 Deploy /api/public/* from backend/src/
- 1.3 pg_dump -> dev RDS restore
- 1.4 CHT dev MEDIAHUB_BASE_URL -> dev API; catalog smoke
- 1.5 Lambdas: post_tagging, playlist_doctor_tagger, cache_clear
- 1.6 CHT: /admin shell with group-gated link

Phase 2: Admin and studio API

- 2.1 /api/admin/studio/* + Cognito JWT middleware
- 2.2 Aurora client_ids sync on login
- 2.3 Port /admin/studio/content + /admin/studio/analytics
- 2.4 Remaining sync jobs -> EventBridge/Lambda

Phase 3: Studio pipeline and prod infra

- 3.1 Port clipper, conversations, render admin pages

- 3.2 Producer Aurora Global prod Terraform
- 3.3 Prod data migration; parallel EC2 48h
- 3.4 Port /admin/studio/hcp-intel/*
- 3.5 DR: ue2 reader, Route53 failover drill

Phase 4: Decommission

- 4.1 Retire EC2 monolith
- 4.2 Retire MediaHub Next.js admin
- 4.3 Retire GoTrue, producer users router
- 4.4 Retire ECS worker if any remains
- 4.5 Post-MVP: mediahub-render GPU + S3

10. Round 1 Dev Exit Criteria

- CHT dev loads catalog from producer dev API (API key)
- HCP upsert round-trip works
- Sync Lambda -> cache clear -> fresh clips visible
- Producer dev RDS populated from EC2 dump
- /admin reachable for chm-admin (studio pages may be stubbed)
- No CHT backend connection to producer Postgres
- End users cannot auth on legacy MediaHub UI

11. Rollback

Phase	Rollback
Dev API	Revert CHT MEDIAHUB_BASE_URL to EC2 prod URL
Prod cutover	Keep EC2 48h; revert DNS/env
Lambdas	Disable EventBridge; re-enable EC2 scheduler temporarily
Aurora	Restore snapshot; keep EC2 DB until sign-off

12. Starter Ticket Backlog

ID	Title	Phase
CH-01	Terraform producer dev RDS + API ECS	1
CH-02	pg_dump restore script + validation	1
CH-03	Lambda: post_tagging, playlist_doctor, cache_clear	1
CH-04	CHT dev env + catalog smoke tests	1
CH-05	Cognito chm-* groups + admin link	0-1

CH-06	/admin layout + guards (CHT repo)	1
CH-07	/api/admin/studio JWT middleware	2
CH-08	Port tag editor + analytics UI	2
CH-09	EventBridge full job catalog	2
CH-10	Producer Aurora Global prod	3
CH-11	Prod cutover + EC2 retire	3-4
CH-12	Port hcp-intel admin UI	3
CH-13	Render GPU worker (post-MVP)	4.5

13. Related Documents

- docs/contenthub-migration-plan.md - markdown source (canonical)
- docs/contenthub-admin-architecture.md - admin routes and group matrix
- docs/cht-public-api-contract.md - CHT catalog endpoints
- docs/cache-sync-contract.md - Lambda to CHT cache clear
- docs/mediahub-platform-cutover.md - prod pg_dump detail
- CHT repo: cht-platform-architecture.pdf, cht-platform-auth.pdf